

Manual for HALFpipe processing of resting-state data

For the ENIGMA-PD fMRI normative modelling project

For questions, feel free to contact us!

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An outline of the secondary proposal is published [here](#).

Please also check out our page on the ENIGMA-PD website [here](#)!

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Brief description of the processing pipeline

The aim of this study is to get a better understanding of the neuranatomical heterogeneity underlying cognitive impairment in PD using normative modelling. Given the involvement of various functional brain circuits in PD-related cognitive impairment that shows high interindividual variety, we aim to develop brain charts of "normal" functional connectivity across the aging lifespan. Using these brain charts, we can compute deviations from normal functional connectivity in a sample of individuals with PD and assess how interindividual variation in deviation from normal connectivity is related to cognitive heterogeneity.

We'll combine the rich and heterogeneous resting-state fMRI data from individuals with Parkinson's disease and healthy controls, with data from the Human Connectome Project Aging (preprocessed in an identical manner with HALFpipe) and UK Biobank (preprocessed using the standard UKB pipeline) to model normative growth curves.

The preprocessing pipeline is for the most part following standard procedures of ENIGMA-fMRI. We will use a second atlas, combining the Schaefer cortical atlas with the [Melbourne Subcortical Atlas](#), which provides more detail about subcortical structures compared with the standard *aseg* atlas. Moreover, we're using a novel version of HALFpipe, which has not formally been released yet (as of Sep-26), which fixes several bugs essential for this preprocessing pipeline.

Manual

Installation

We're using a new version of HALFPipe that deals with bugs essential for resting-state fMRI preprocessing. **Please, do not use the standard "latest" HALFPipe version for this project.**

- Apptainer/Singularity: download the HALFPipe .sif image [here](#). Transfer it to your analysis environment/HPC cluster.
- **Docker: Instructions will follow soon!**

For more information on the installation of HALFPipe, please refer to <https://fmri.science/halfpipe/installation.html>.

Note that you'll also need a FreeSurfer license for this preprocessing pipeline. Most ENIGMA-PD sites will have one; otherwise, you can request one [here](#).

Download prerequisite atlas and seed images

We'll use two atlases for the resting-state functional connectivity matrices (download [here](#)): the HALFPipe standard **Schaefer 400P17N + aseg atlas** ("Schaefer2018Combined") and the **Schaefer 400P7N + Melbourne Subcortical Atlas S3** ("Schaefer400P7NMSAS3"). Store these in your working directory, in a folder named e.g., atlases.

Additionally, we would like to use this opportunity to also compute seed-based connectivity of the substantia nigra and nucleus basalis of Meynert (download [here](#)). Store these in your working directory, in a folder named e.g., seed_regions.

Prepare your data

- **Is your data structured in BIDS format?** Fantastic! You can proceed to preprocessing.
- **Is your data not structured in BIDS format?** No worries. Just make sure data is organized in a structured manner, so that participant identifiers (e.g., sub-0001) can be replaced with wildcards ({subject}). The same holds for {session}, {run}, {acquisition}, and {task} identifiers. More information can be found [here](#) (see tab "If not in BIDS").

Prepare and run HALFPipe processing using the HALFPipe GUI

For this project, we'll deviate from standard ENIGMA fMRI preprocessing in using an additional atlas ("Schaefer400P7NMSAS3"). For the rest, we'll follow the standard HALFPipe preprocessing pipelines, which are outlined in detail [here](#).

Note that it might be advisable to first run HALFPipe for 2-3 participants, to assess if everything is working as expected. Running the pipeline for a large sample requires significant memory; even the preparatory steps outlined below may take a while to prepare.

- **Launch HALFPipe** as follows:
 - o Apptainer/Singularity:
apptainer run --contain --cleanenv --bind /:/ext halfpipe-1.3.3.dev91+g892515660.sif --use-cluster --tui
 - o **Docker:**
Instructions will follow soon!
- Follow the instructions under **"Pipeline settings"** as described in the HALFPipe manual.

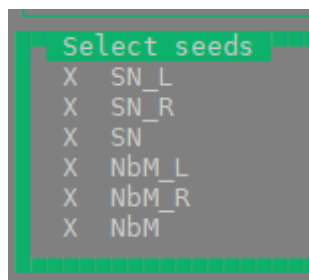
- For **features**, we'll use section 1 (seed-based connectivity) and 2 (atlas-based connectivity matrix)

- **Seed-based connectivity:** follow the [instructions in the HALFpipe manual](#).

For the seed images, use the supplied seed images (in your seed_regions folder) of the substantia nigra and nucleus basalis of Meynert:

- SN_L.nii.gz, enter with label SN_L
- SN_R.nii.gz, enter with label SN_R
- SN_Bilateral.nii.gz, enter with label SN
- NBM_L.nii.gz, enter with label NbM_L
- NBM_R.nii.gz, enter with label NbM_R
- NBM_Bilateral.nii.gz, enter with label NbM

A list of six seeds should be selectable:



Use the five denoising strategies that are outlined in [the HALFpipe manual](#):

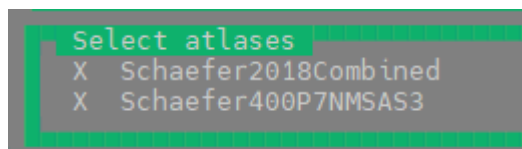
- 1) aCompCor
- 2) Motion parameters with scrubbing
- 3) Pipeline 2 + Global Signal (GSR)
- 4) Motion parameters
- 5) Pipeline 4 + Global Signal (GSR)

- **Atlas-based connectivity matrices:** again, follow the instructions in the HALFpipe manual.

For the atlas images use the supplied seed images (in your atlases folder) of the two different atlases:

- atlas-Schaefer2018Combined_dseg.nii.gz, enter with label Schaefer2018Combined
- atlas-Schaefer400P7N-MSA_dseg.nii.gz, enter with label Schaefer400P7NMSAS3

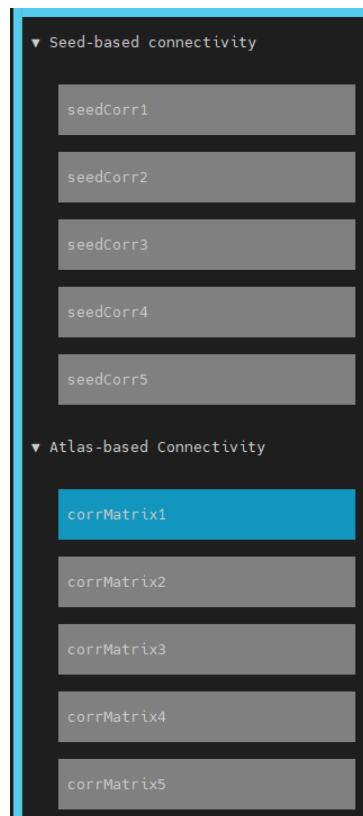
A list of two atlases should be selectable:



Again, use the five denoising strategies that are outlined in [the HALFpipe manual](#):

- 1) aCompCor
- 2) Motion parameters with scrubbing
- 3) Pipeline 2 + Global Signal (GSR)
- 4) Motion parameters
- 5) Pipeline 4 + Global Signal (GSR)

In the end, the **features** tab should have ten fields: five seed-based connectivity fields, and five correlation matrix fields:



▼ Seed-based connectivity

seedCorr1

seedCorr2

seedCorr3

seedCorr4

seedCorr5

▼ Atlas-based Connectivity

corrMatrix1

corrMatrix2

corrMatrix3

corrMatrix4

corrMatrix5

- You can skip the “Group level models” tab
- Under “Check and run”, a json file should be visible that looks like [this example spec.json](#).

Note that when running HALFpipe on an HPC, HALFpipe will produce files to submit jobs on the cluster, like sbatch scripts (submit.sge.sh, submit.slurm.sh, submit.torque.sh). Please note that some slurm settings (time, memory) should be adapted and some settings should be added, dependent on the HPC you’re working on.

Finished?

Quality control

Instructions on quality control, with examples of good and bad quality, are reported in [the HALFpipe manual](#). In our experience, especially the ICA AROMA may be difficult to assess. Please, let us know if you experience any problems with quality control that we may be able to assist in.

Please register the framewise displacement (FD), DVARS and other potential image quality measures for your subjects, as these will be used in the analyses.

Ready to share your data?

That’s great! Please let us know, and we’ll send further instructions.